BLK II - UNIT 3: Weather Prediction Models
b. Model Interpretation

Our course has emphasized weather analysis so far. Accurate forecasting depends on an accurate understanding of the current state of the atmosphere and what meteorological processes are affecting certain features. The previous unit introduced you to the concept of model data, and the importance of verifying the accuracy of these computer-generated forecasts. This unit will now explore various models that have specific uses and are used to gather specific information. Many of these models will look very similar to some of the tools you have used in the analysis portion of the course. Now we will shift our focus from the charts displaying current data to these model charts predicting the future state of the atmosphere.

The goal of this objective will be to show you have the capability to interpret a wide variety of weather data from several different model products. To successfully demonstrate competency in this topic, you will be given information through guided lectures, as well as a series of questions and tasks designed to display your ability to correctly interpret each model product discussed throughout this objective. These foundational forecasting concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instructions for completing all requirements necessary to successfully complete this objective.

INSERT SOP HERE

Constant Pressure Model Charts

You have already covered how to interpret weather features using constant pressure charts. We will be utilizing the same standard pressure levels to interpret the future intensity, and location of significant weather features. Key considerations when reading these charts are to be able to identify the main features and to identify any movement or intensity changes. Like analysis charts, these model charts are used to:

- Identify weather systems aloft
- Assess atmospheric stability
- Determine steering winds for surface features
- Infer areas of potential severe weather conditions

These charts still utilize the same structure of lines of equal value that you should be familiar with at this point.

- Isobars (Lines of constant pressure): Indicate pressure patterns, revealing high- and low-pressure centers. Tighter packing means stronger pressure gradient and stronger winds.
- Height Contours: Show the height of the atmosphere above the surface.
- Winds: Wind barbs show wind speed and direction. Use the winds to see how air is moving around the pressure systems.
- Temperature: Isotherms are used to locate areas of warm or cold air advection.

1000-500mb Thickness Model

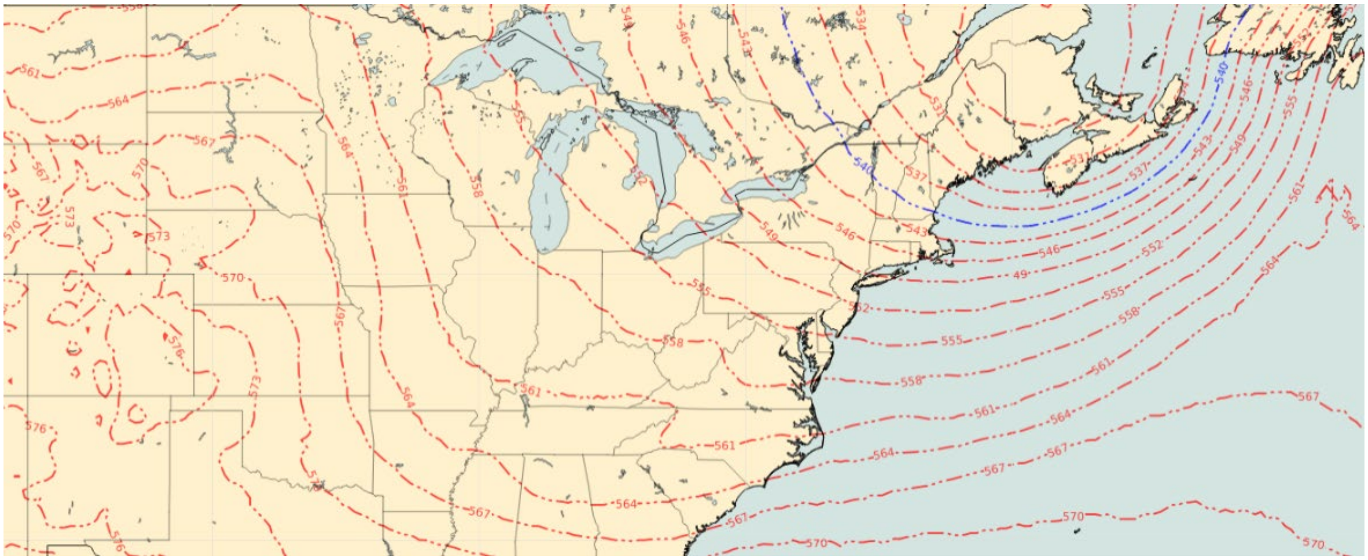
Thickness is the vertical distance between two constant pressure surfaces. The distance between 1,000 and 500 mb quantifies the dynamic atmosphere. Forecast models assume the surface is 1,000 mb. Forecast models use thickness to depict the average temperature of a layer. High thickness values indicate warmer air while low thickness values indicate colder air. Thickness isopleths are lines of equal thickness like isotherms are lines of equal temperature. The thickness lines combine to reveal a gradient separating warm and cold air that will resemble temperature gradients.

Thickness gradients depict areas of thermal advection. Models depict thermal advection in areas with a tight thermal (thickness) gradient and a layer's isobaric or contour gradient. You find advection by determining the wind flow shown by the isobaric gradient. When two successive isobars intersect two successive thickness lines, they form an advection box. The box's size indicates the advection's intensity. Large boxes are weak advection while small boxes are strong advection. Wind flow also determines the type of advection. Flow from higher to lower thickness values indicate a southerly flow and equates to warm air advection (WAA). This is common on the east side of low-pressure centers. Flow from lower to higher thickness values indicate a northerly flow and equates to cold air advection (CAA) which is most common on the east side of high-pressure centers.

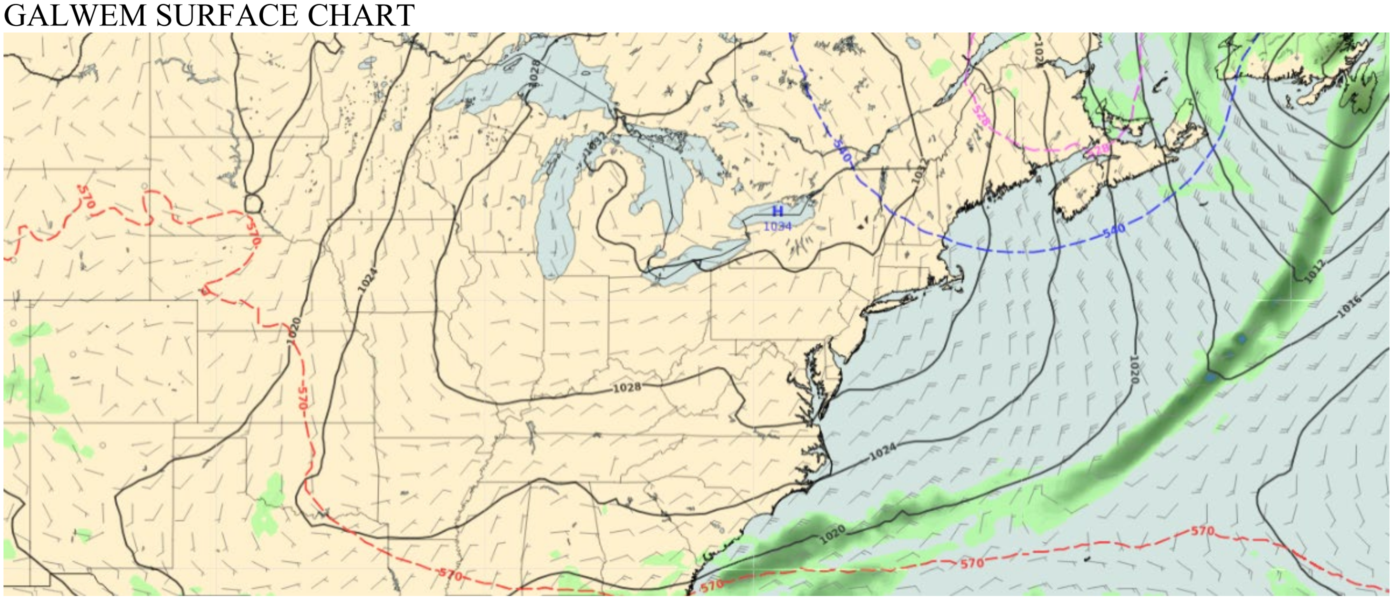
Models infer the location of the Polar front jet centered within the tightest thickness gradient, paralleling the 540 decameter (dam) thickness line. The thickness lines also infer air mass boundaries. The 528-dam line is the boundary between arctic and polar air representing the edge of the Arctic high meaning values less than 528-dam is arctic air. The 540-dam line is the rain/snow delineation line and represents the boundary between polar and modified-polar air. The 570-dam line is the boundary between modified polar and tropical air. Values greater than 570 dam is tropical air.

Lastly, warm and cold fronts are on the warm side of the tightest thickness (thermal) gradient. Warm air advection is poleward of warm fronts and cold air advection is poleward of cold fronts. Cold and Warm Fronts always lie in pressure troughs either extending from the low-pressure center or from the triple point if it is a mature or decaying wave. Occluded fronts will extend from the triple point through the thickness (thermal) ridge on the northeast side of the low-pressure center.

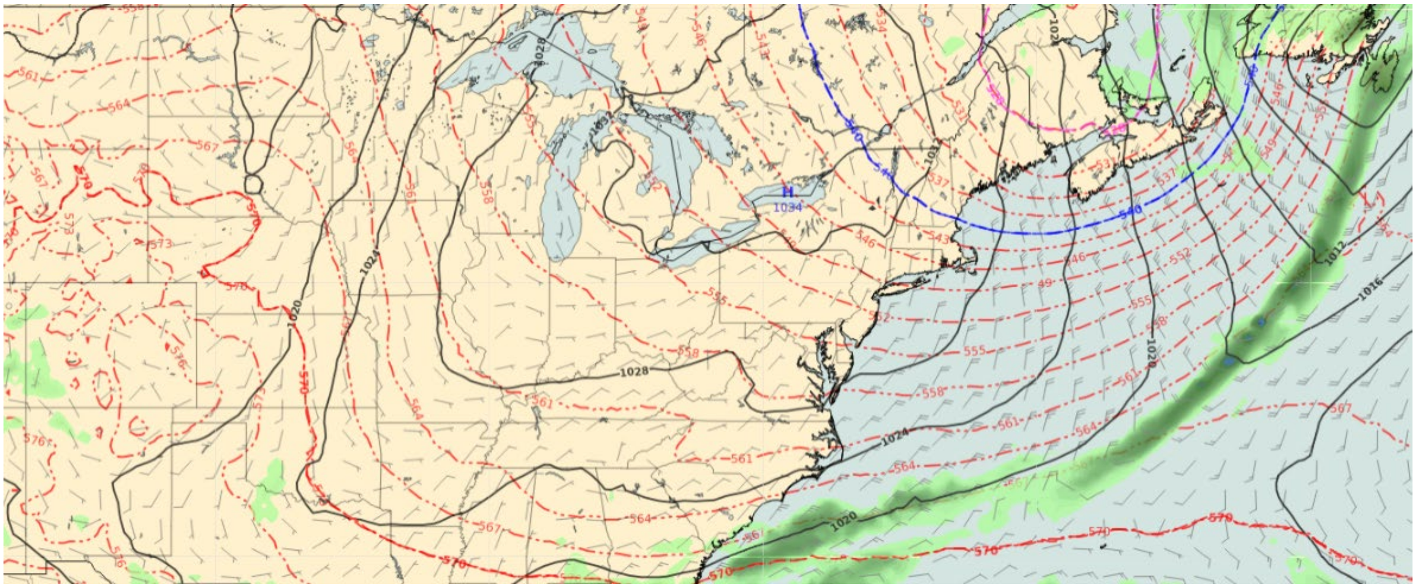
GALWEM THICKNESS CHART



GALWEM SURFACE CHART



GALWEM COMBINED CHART

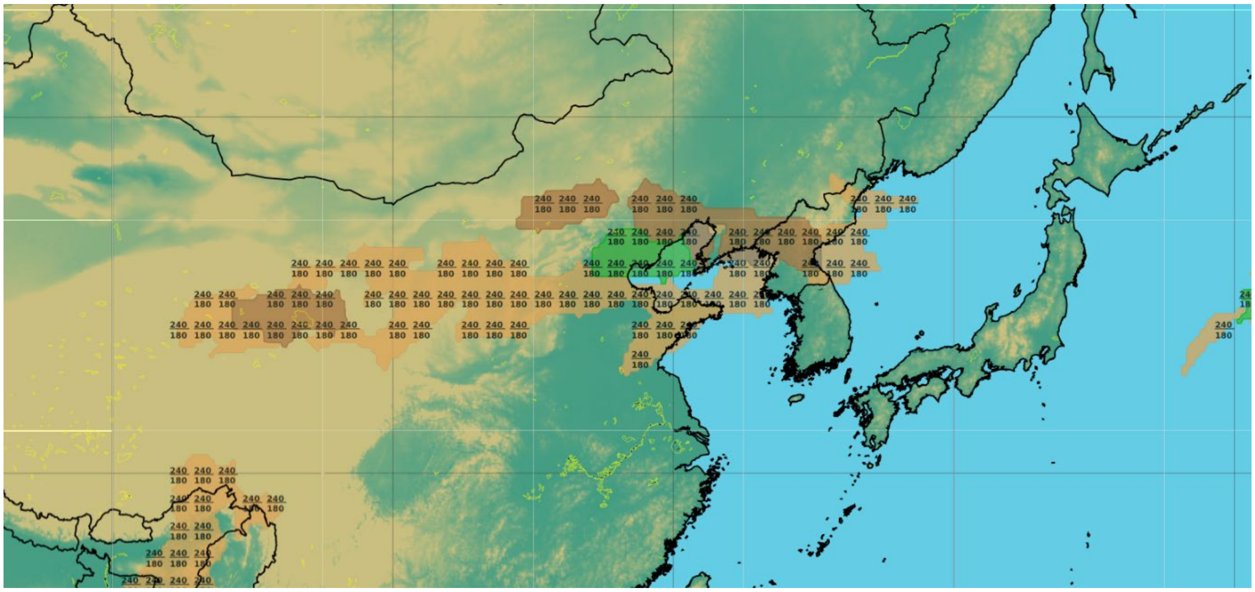


Practice: 1000-500mb Thickness Model

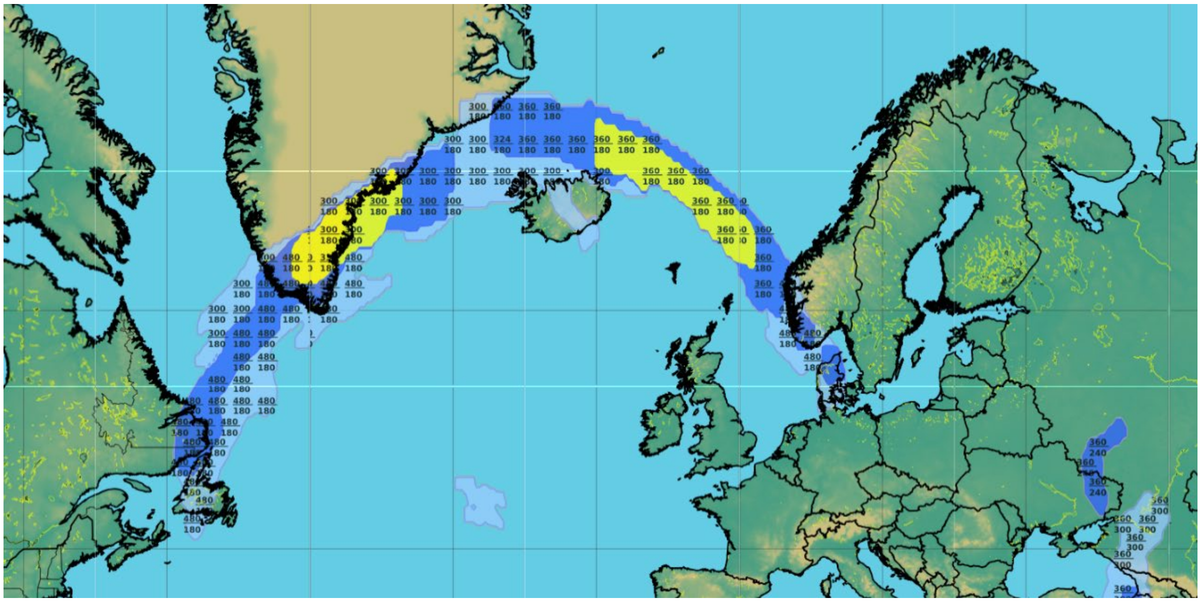
Hazard Models (Icing, Turbulence, Thunderstorm)

Hazard Models are forecast charts that predict the likelihood and intensity of aviation hazards: Icing, Turbulence, and Thunderstorms. Specific models will vary by forecast center. These models are crucial for identifying Potential Mission Hazards.

Icing Charts: Predict areas with potential for ice accumulation on aircraft. Helps determine severity (e.g., light, moderate, severe), type of icing (e.g., clear, rime, mixed), and altitude levels where icing is most likely.



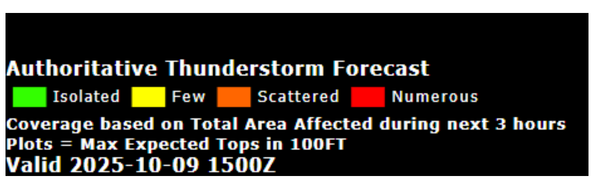
Turbulence Charts: Forecast regions of atmospheric instability, causing bumpy or hazardous flight conditions. Pinpoint severity (e.g., light, moderate, severe) and altitude levels of turbulence.

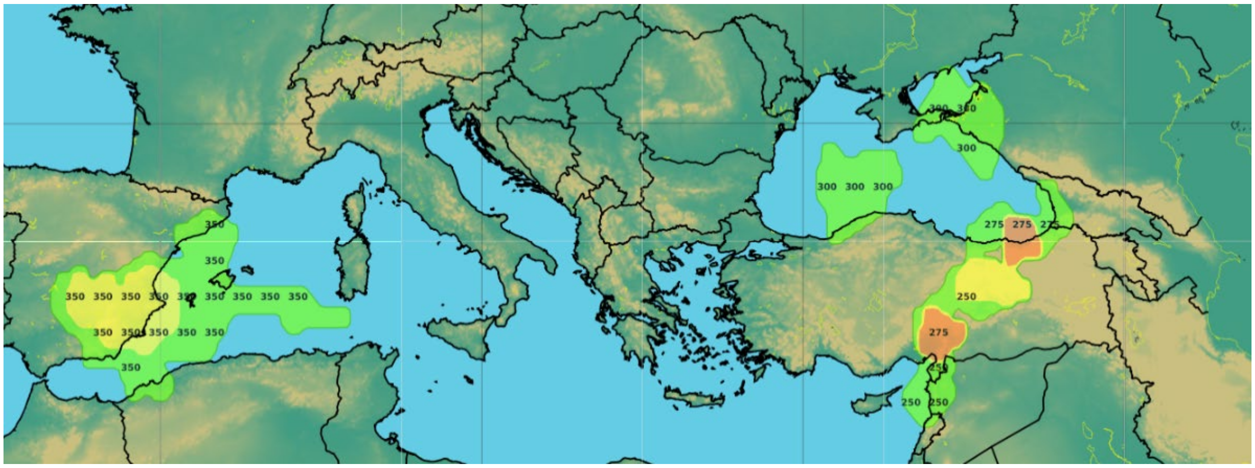


Thunderstorms Charts: Identify areas with the potential for thunderstorms. Determines coverage (isolated, scattered, numerous) and maximum thunderstorm cloud top heights. Justify the potential by examining Skew-T indices.

The basics of how to read these tools:

- Identify Hazardous Conditions: Locate hazard charts from the appropriate forecast center.
- Specific Annotations: If hazards are identified for the location:
- Thunderstorms: Note the coverage (isolated, scattered, numerous) and maximum cloud tops.
- Turbulence & Icing: Annotate the severity (light, moderate, severe), type, and altitude levels (e.g., FL200-FL250).



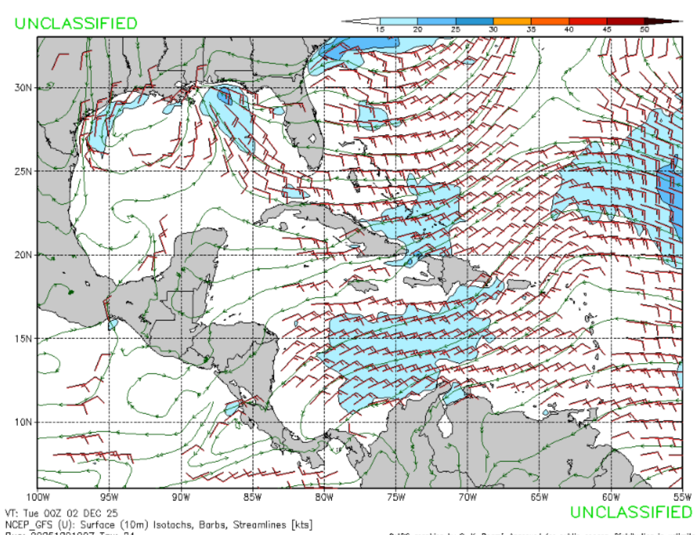


Practice: Hazard Models

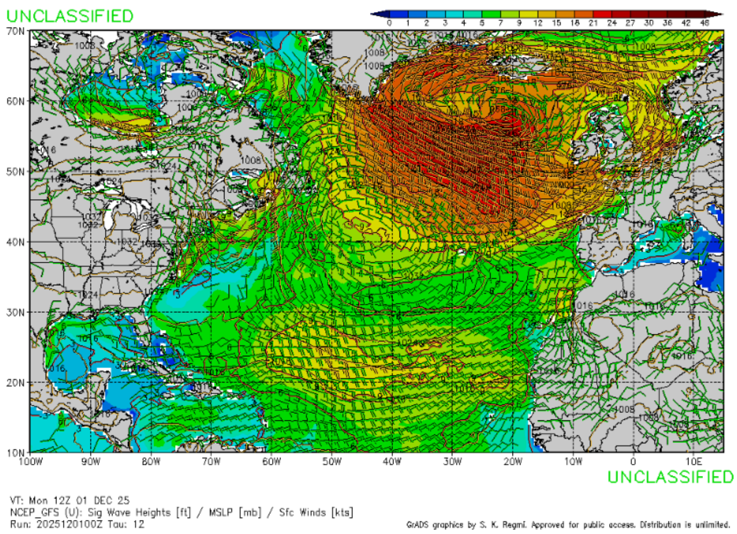
FNMOC Models

WxMap

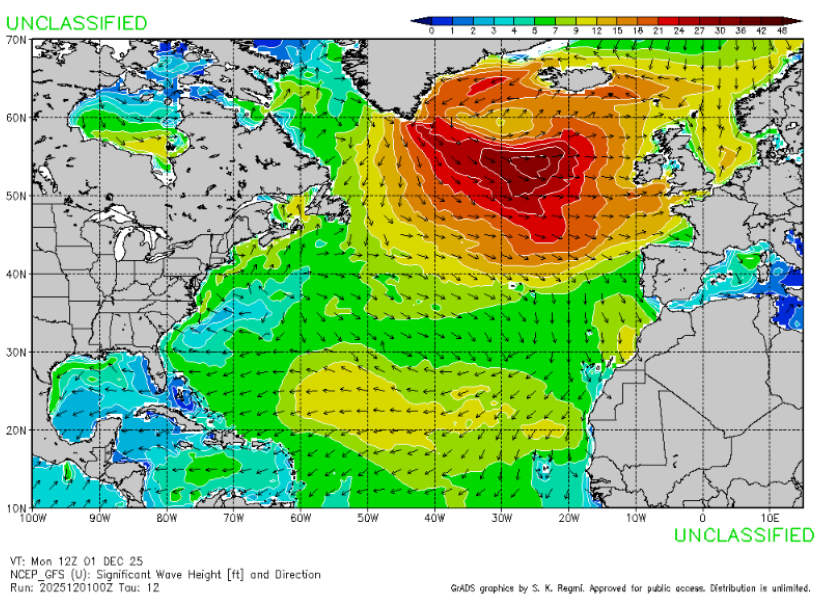
Surface (10 meter) Wind Barbs, Streamlines, Isotachs [kts]: Used for analyzing wind patterns and weather systems. The wind barbs act as a representative symbol of wind speed with increments of 5 knots (KT). The streamlines identify the flow of the wind directions, helping to identify various weather features. The isotachs are lines that represent the flow with a constant wind speed. They are used to visualize the horizontal flow of air at a given level and for this model it would be the surface or rather features within ten meters of the surface.



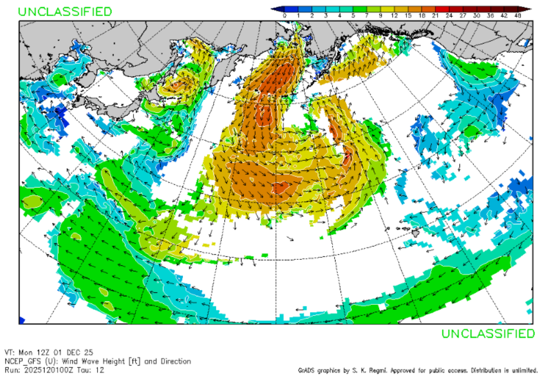
Sig Wave Heights [ft], Over Ocean Sfc Winds [kt]: This model depicts the Significant Wave height in association with Surface winds. The significant wave height is the highest third of waves in open waters and is measured from trough to crest. This model aids in observing both meteorological and atmospheric features. Furthermore, it shows the relationship between wind speed and fetch area has on winds as faster winds impart more energy to the waves, increasing wave amplitude (height), while longer fetch areas yield higher waves.



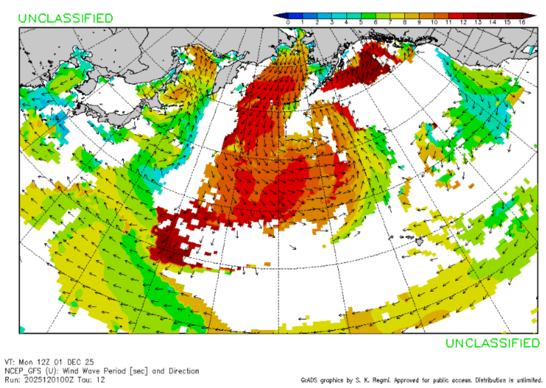
Significant Wave Height & Direction: Significant wave height depicts the highest third of waves in open waters, the most impactful waves in a sea state, while wave direction captures the prevailing direction of wave propagation. The knowledge of the significant waves and the direction in which they are flowing aids in forecasting and maritime risk management acting as a predictor for sea conditions to aid in navigation and offshore operations.



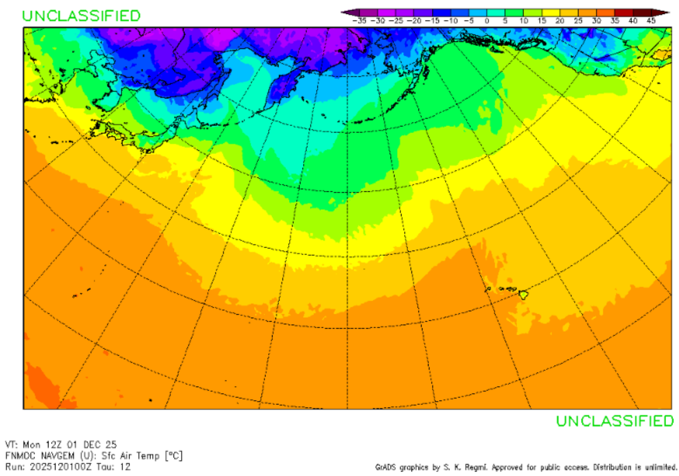
Wind Wave Height & Direction: This model depicts waves generated from surface wind conditions giving the average direction and heights associated with the waves. It aids in the identification in fetch areas and the intensification of waves associated with intense atmospheric conditions.



Wind Wave Period & Direction: Wave periods are the time it takes for two successive wave crests to pass a given point. There is a relationship between wave period and wave amplitude. Generally, a wave with a large wave amplitude will have a longer wave period, and a wave with a smaller wave amplitude will have a shorter wave period. Your model charts on FNMOC, which will show period and direction, but this should also be initialized and verified with life soundings such as buoys or trained observers.

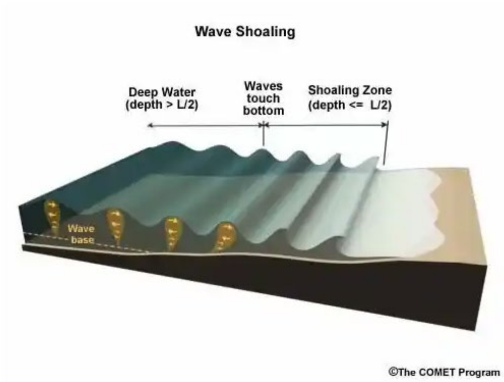


Sea Surface Temperature: Sea surface temperature is the representative model for the water near the surface of the ocean ranging from the surface down to about 5 meters. It has many military uses regarding Anti-Submarine Warfare, Naval and Surface Fleet Operations, Atmospheric considerations, and water survivability for personnel.



Littoral Zone Considerations

Sig. Breaker Heights: Significant breaker height is normally used for the surf zone. The most important characteristics of the surf zone near any beach landing site are the significant breaker height, breaker period, and breaker type. Determining significant breaker heights requires significant wave heights, predominant swell waves, affecting a given landing zone. This can be determined by buoys, manual observers, or verified models. Then taking the swell period, you compare it to your hand chart to get the shoaling factor, the measure of how waves increase in height as they enter shallower water. Take the shoaling factor and multiple it by your significant wave heights to receive your significant breaker heights.



Sea Surface Temperature: Sea Surface Temperature is the temperature of ocean water near the surface. The Sea surface Temperature has impacts on both personnel and the atmosphere. For ship to shore operations, it can greatly impact personnel as they make approach to shore if for some reason their vessel capsizes. At this point understanding sea surface temperature can affect the survivability in the water as the temperature of the water can cause personnel to be in hypothermic conditions. To determine Sea surface temperature the use of microwave or infrared satellites can depict the temperature of the water, buoy and ships with temperature sensors can produce observations of the temperature of the water, and verified models can help aid in the establishment of your sea surface temperature.

Shoreline Orientation: Beaches are irregular in shape and structure, understanding that terrain features of a beach affect breaker heights within the surf zone can determine go versus no-go criteria for amphibious landings. Focal points are convex portions of a beach; portions of the beach rounded outward into the water. These focal points intensify the energy of breaker waves as the wave coalesce together around the Focal energy, sharing their energy to increase the height of the breaker waves. This is different from a straight beach as energy is spread evenly across the beach. This is also different than concave shapes of the beach, as the inward shape of an object creates barriers on its sides allowing for the waves to break across the focal points creating that shadow effect allowing for lower wave heights to occur. To forecast for the difference between focal points, concave points, and straight beaches requires the forecaster to understand the orientation of the land they are forecasting for. The use of satellite imagery is needed and the understanding of the predominate wave direction.


FNMOC

- **CTG 80.7 CC**
- **Ocean Products**
    - **Sea Height / Swell:**
    - **Sea Surface Temps:**
    - **High Winds / Sea Warnings:**
    - **Currents:**

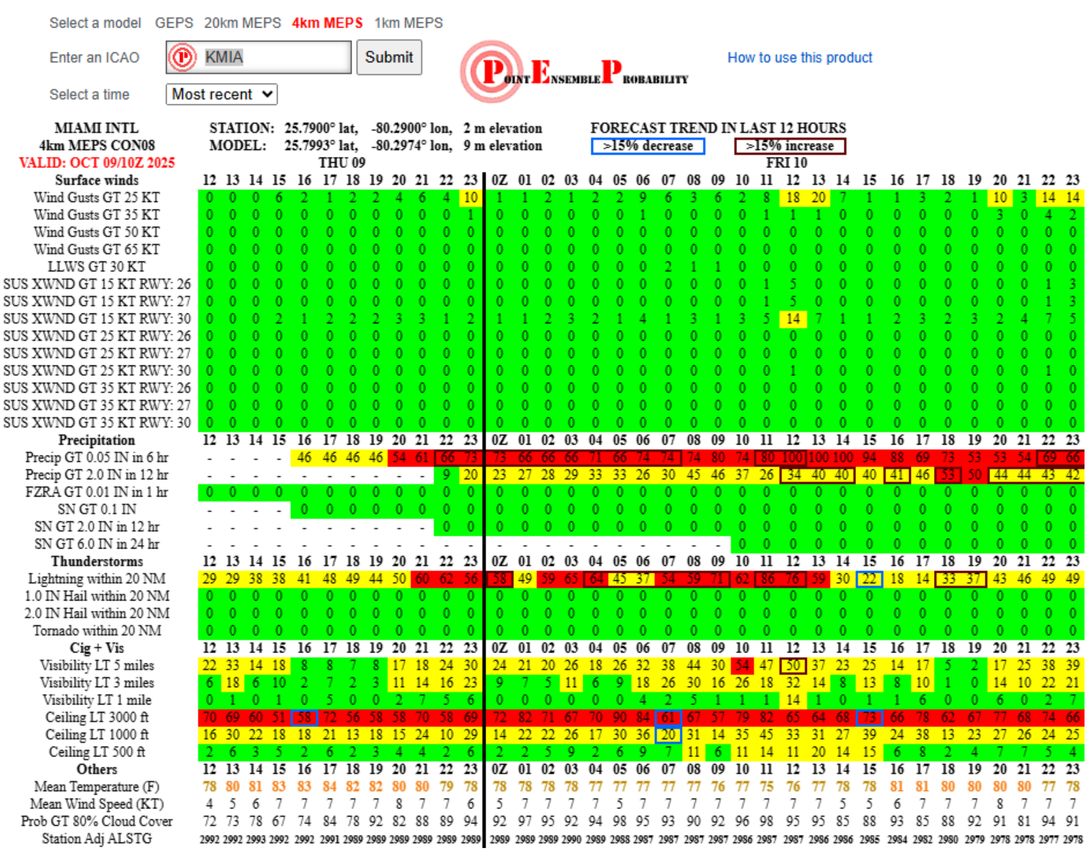
Practice: FNMOC Models

<u>Model SKEW-T</u>

Skew-T diagrams are extremely useful when analyzing severe weather parameters and to justify the intensity of severe weather and other potential hazards. Like the constant pressure charts, a model Skew-T will read exactly like an analysis Skew-T. The only difference will be the necessary verification process to determine the accuracy of the data on the model. If the model Skew-T has an inaccurate verification, this could drastically change the intensity, timing, and type of weather features indicated in the indices.

Point Ensemble Probability Bulletin (PEP)

Point Ensemble Probability (PEP) bulletins are numerical guidance based on the probability of various weather phenomena. Computed forecast values are from two sources: the Global Ensemble Prediction System (GEPS) and the Mesoscale Ensemble Prediction System (MEPS). Yellow indicates a greater than 10% chance of a weather parameter (e.g., wind speed, precipitation) exceeding a specified threshold. Red indicates a greater than 50% chance of exceeding the threshold.



Select a model GEPS 20km MEPS 4km MEPS 1km MEPS

Enter an ICAO KMIA Submit

Select a time Most recent

POINT ENSEMBLE PROBABILITY

How to use this product

MIAMI INTL STATION: 25.7900° lat, -80.2900° lon, 2 m elevation

4km MEPS CON08 MODEL: 25.7993° lat, -80.2974° lon, 9 m elevation

VALID: OCT 09/10Z 2025

FORECAST TREND IN LAST 12 HOURS: >15% decrease >15% increase

THU 09	12	13	14	15	16	17	18	19	20	21	22	23
Surface winds	12	13	14	15	16	17	18	19	20	21	22	23
Wind Gusts GT 25 KT	0	0	0	6	2	1	2	2	4	6	4	10
Wind Gusts GT 35 KT	0	0	0	0	0	0	0	0	0	0	0	1
Wind Gusts GT 50 KT	0	0	0	0	0	0	0	0	0	0	0	0
Wind Gusts GT 65 KT	0	0	0	0	0	0	0	0	0	0	0	0
LLWS GT 30 KT	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 15 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 15 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 15 KT RWY: 30	0	0	0	2	1	2	2	2	3	3	1	2
SUS XWND GT 25 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 25 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 25 KT RWY: 30	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 35 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 35 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 35 KT RWY: 30	0	0	0	0	0	0	0	0	0	0	0	0
Precipitation	12	13	14	15	16	17	18	19	20	21	22	23
Precip GT 0.05 IN in 6 hr	-	-	-	-	46	46	46	46	54	61	66	73
Precip GT 2.0 IN in 12 hr	-	-	-	-	-	-	-	-	-	-	9	20
FZRA GT 0.01 IN in 1 hr	0	0	0	0	0	0	0	0	0	0	0	0
SN GT 0.1 IN	-	-	-	-	0	0	0	0	0	0	0	0
SN GT 2.0 IN in 12 hr	-	-	-	-	-	-	-	-	-	-	0	0
SN GT 6.0 IN in 24 hr	-	-	-	-	-	-	-	-	-	-	-	-
Thunderstorms	12	13	14	15	16	17	18	19	20	21	22	23
Lightning within 20 NM	29	29	38	38	41	48	49	44	50	60	62	56
1.0 IN Hail within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0
2.0 IN Hail within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0
Tornado within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0
Cig + Vis	12	13	14	15	16	17	18	19	20	21	22	23
Visibility LT 5 miles	22	33	14	18	8	8	7	8	17	18	24	30
Visibility LT 3 miles	6	18	6	10	2	7	2	3	11	14	16	23
Visibility LT 1 mile	0	1	0	1	0	5	0	0	2	7	5	6
Ceiling LT 3000 ft	70	69	60	51	58	72	56	58	58	70	58	69
Ceiling LT 1000 ft	16	30	22	18	18	21	13	18	15	24	10	29
Ceiling LT 500 ft	2	6	3	5	2	6	2	3	4	4	2	6
Others	12	13	14	15	16	17	18	19	20	21	22	23
Mean Temperature (F)	78	80	81	83	83	84	82	82	80	80	79	78
Mean Wind Speed (KT)	4	5	6	7	7	7	7	7	8	7	7	6
Prob GT 80% Cloud Cover	72	73	78	67	74	84	78	92	82	88	89	94
Station Adj ALSTG	2992	2992	2993	2992	2992	2991	2989	2989	2989	2989	2989	2989

FRI 10	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Surface winds	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Wind Gusts GT 25 KT	1	1	2	1	2	2	9	6	3	6	2	8	18	20	7	1	1	3	2	1	10	3	14	14
Wind Gusts GT 35 KT	0	0	0	0	0	0	1	0	0	0	0	1	1	1	0	0	0	0	0	0	3	0	4	2
Wind Gusts GT 50 KT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind Gusts GT 65 KT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LLWS GT 30 KT	0	0	0	0	0	0	0	2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 15 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	1	5	0	0	0	0	0	0	0	0	0	1	3
SUS XWND GT 15 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	1	5	0	0	0	0	0	0	0	0	0	1	3
SUS XWND GT 15 KT RWY: 30	1	1	2	3	2	1	4	1	3	1	3	5	14	7	1	1	2	3	2	3	2	4	7	5
SUS XWND GT 25 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 25 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 25 KT RWY: 30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0
SUS XWND GT 35 KT RWY: 26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 35 KT RWY: 27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUS XWND GT 35 KT RWY: 30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Precipitation	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Precip GT 0.05 IN in 6 hr	73	66	66	66	71	66	74	74	74	80	74	80	100	100	100	94	88	69	73	53	53	54	69	66
Precip GT 2.0 IN in 12 hr	23	27	28	29	33	33	26	30	45	46	37	26	34	40	40	40	41	46	53	50	44	44	43	42
FZRA GT 0.01 IN in 1 hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SN GT 0.1 IN	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SN GT 2.0 IN in 12 hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SN GT 6.0 IN in 24 hr	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Thunderstorms	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Lightning within 20 NM	58	49	59	65	64	45	37	54	59	71	62	86	76	59	30	22	18	14	33	37	43	46	49	49
1.0 IN Hail within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2.0 IN Hail within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Tornado within 20 NM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cig + Vis	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Visibility LT 5 miles	24	21	20	26	18	26	32	38	44	30	54	47	50	37	23	25	14	17	5	2	17	25	38	39
Visibility LT 3 miles	9	7	5	11	6	9	18	26	30	16	26	18	32	14	8	13	8	10	1	0	14	10	22	21
Visibility LT 1 mile	0	0	0	0	0	0	4	2	5	1	1	1	14	1	0	1	1	6	0	0	6	0	2	7
Ceiling LT 3000 ft	72	82	71	67	70	90	84	61	67	57	79	82	65	64	68	73	66	78	62	67	77	68	74	66
Ceiling LT 1000 ft	14	22	22	26	17	30	36	20	31	14	35	45	33	31	27	39	24	38	13	23	27	26	24	25
Ceiling LT 500 ft	2	2	5	9	2	6	9	7	11	6	11	14	11	20	14	15	6	8	2	4	7	7	5	4
Others	0Z	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Mean Temperature (F)	78	78	78	78	77	77	77	77	77	76	77	75	76	77	78	78	81	81	80	80	80	80	77	78
Mean Wind Speed (KT)	5	7	7	7	7	5	7	7	7	7	7	7	7	7	5	5	6	7	7	7	8	7	7	7
Prob GT 80% Cloud Cover	92	97	95	92	94	98	95	93	90	92	96	98	95	95	85	88	93	85	88	92	91	81	94	91
Station Adj ALSTG	2989	2989	2989	2990	2989	2988	2987	2987	2987	2987	2986	2987	2987	2986	2986	2985	2984	2982	2980	2979	2978	2978	2977	2978

Surface winds	Greater than (GT) or equal to 25 knots
	GT or equal to 35 knots
	GT or equal to 50 knots
Precipitation	6-hr Precipitation GT 0.10 in
	12-hr Precipitation GT 2.0 in
	Freezing rain GT 0.01 in
	6-hr Snow Accumulation GT 0.1 in
	2-hr Snow Accumulation GT 2.0 in
	24-hr Snow Accumulation GT 6.0 in
Thunderstorms	Lightning strike
	Hail GT or equal to 0.75 in
Ceiling and visibility	Visibility less than (LT) 5 miles
	Visibility LT 3 miles
	Visibility LT 1 mile
	Ceilings below 3,000 ft
	Ceilings below 1,000 ft
Mean temperature	Fahrenheit degrees
Mean wind speed	Knots

Exercise / Practice: PEP Model

Meteogram

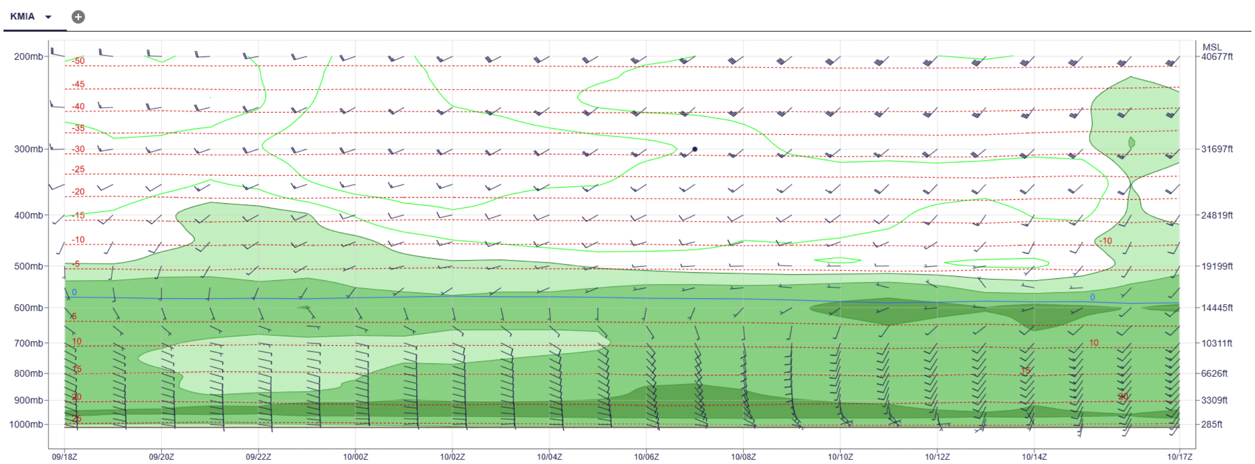
Meteograms display meteorological variables plotted against a time axis, allowing users to observe trends, correlations, and predict future weather events. While the specific variables included and their representation may vary, the underlying principles of interpretation remain consistent.

Analyzing a meteogram involves several key steps:

- **Trend Identification:** Observing the changes in individual variables over time (e.g., increasing, decreasing, or fluctuating temperatures).
- **Correlation Analysis:** Examining the relationships between different variables (e.g., how temperature changes in relation to cloud cover).
- **Event Prediction:** Using observed trends and correlations to forecast future weather events (e.g., predicting the onset and duration of precipitation).
- **Air Advection Assessment:** Evaluating warm or cold air advection by observing changes in temperature and 1000-500mb Thickness. An increasing temperature or 1000-500mb Thickness suggests warm air advection, while decreasing values indicate cold air advection.

It is important to note that meteograms provide forecasts for a single location and their accuracy is contingent on the quality of the underlying weather model and input data. Users should be aware of the potential for inherent errors and uncertainties in the forecasts. This section will equip the user with the knowledge to effectively interpret meteograms, while remaining cognizant of their limitations.

Time Axis: The horizontal axis represents time, typically spanning hours to days. Time progresses from left to right.



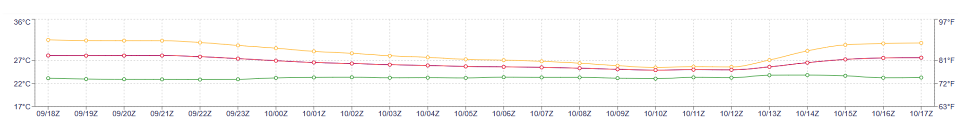
Meteorological Variables: Variables are represented on the vertical axis. Common variables and their typical representations are described below. Note that the specific variables and their units should be clearly labeled on the meteogram. Temperature (T): Displayed as a line graph. Units are typically in degrees Celsius (°C) or Fahrenheit (°F). Precipitation (P): Often represented as vertical bars, indicating the amount of liquid-equivalent precipitation (rain, snow, etc.) accumulated over a given time interval. Units are typically millimeters (mm) or inches (in). Probability of precipitation is sometimes shown.

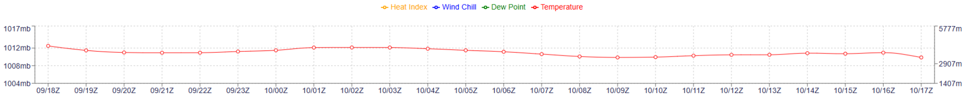
Wind: Wind speed and direction are displayed using wind barbs or arrows. Wind direction indicates the direction from which the wind is blowing. Wind speed is indicated by the number of barbs or the length of the arrow. Units for wind speed are typically knots (kt).

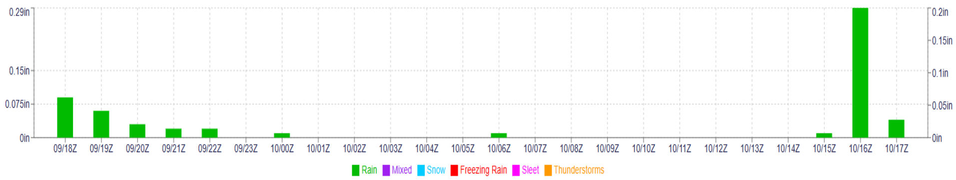
Cloud Cover: Depicted using shading or symbols representing the fraction of the sky covered by clouds.

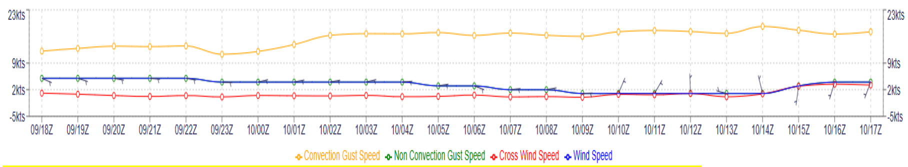
Pressure (p): Displayed as a line graph. Units are typically in hectopascals (hPa) or millibars (mb).

Humidity (RH): Represented as a line graph typically as a percentage (%).









Exercise / Practice: Meteogram Model

TARP

The TARP bulletin presents numerical weather model output tailored to assess potential weather-related risks and inform proactive resource management strategies. This document will equip users with the necessary knowledge to effectively extract and apply the data presented in the TARP bulletin, enabling informed decision-making.

The TARP Numerical Weather Bulletin presents a forecast of weather conditions within a specific time frame with the goal of weather-related risk assessment for effective resource allocation. The bulletin is presented in a structured numerical output format.

Accurate interpretation requires careful assessment of several factors:

- Forecast Horizons: Understanding the temporal range of the forecast data and the increasing uncertainty associated with longer-range projections.
- Variable Thresholds: Recognizing pre-defined thresholds for specific variables that trigger potential threat levels (e.g., wind speed exceeding a critical limit for infrastructure stability, precipitation levels leading to flood risk).
- Scenario Analysis: Combining multiple weather variables to assess complex, compound threat scenarios (e.g., high winds combined with freezing rain creating significant ice accumulation risks).
- Risk Mitigation Strategies: Utilizing the information to select and implement appropriate mitigation strategies to minimize the potential adverse impacts.

KBIX - KEESLER AFB - GALWEM 2025-10-09T06:00:00Z Extracted Model Run

Valid	0 hr Thu 09/06Z 09/01L	1 hr Thu 09/07Z 09/02L	2 hr Thu 09/08Z 09/03L	3 hr Thu 09/09Z 09/04L	4 hr Thu 09/10Z 09/05L	5 hr Thu 09/11Z 09/06L	6 hr Thu 09/12Z 09/07L	7 hr Thu 09/13Z 09/08L	8 hr Thu 09/14Z 09/09L
Temp °F	77°F	76°F	75°F	75°F	74°F	73°F	73°F	73°F	74°F
Temp °C	25°C	24°C	24°C	24°C	23°C	23°C	23°C	23°C	23°C
DP Temp °F	69	67	67	67	67	67	67	68	68
DP Temp °C	20	20	20	20	20	20	20	20	20
Humidity	76%	75%	76%	78%	79%	81%	84%	84%	81%
Wind Chill	77°F	76°F	75°F	75°F	74°F	73°F	73°F	73°F	74°F
Heat Index	78	77	76	76	75	73	73	73	74
FITS	91	90	89	89	88	88	87	87	88
Gen Wx	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon
Cloud Coverage	8	8	8	8	8	8	6	2	1
Lowest Ceiling	051	059	059	059	059	076	076		
Winds (kt)	020/11 19	020/11 19	020/11 19	020/12 20	030/12 20	030/12 19	020/11 19	020/11 19	030/12 22
Cross Wind (kt)	3	2	1	1	1	1	1	1	1
1,000 ft MSL Winds	060/24	060/24	060/24	060/23	060/23	060/22	060/21	060/19	060/19
2,000 ft MSL Winds	070/22	070/21	070/19	070/18	070/16	070/16	060/15	050/14	050/15
3,000 ft MSL Winds	080/19	080/17	080/14	080/12	070/10	060/10	040/12	040/13	040/14
4,000 ft MSL Winds	080/13	090/11	090/9	070/7	050/7	040/8	030/11	030/14	030/15
5,000 ft MSL Winds	080/9	090/7	080/6	050/6	030/8	030/10	030/13	030/15	030/17
6,000 ft MSL Winds	080/6	080/5	060/4	030/6	030/10	030/12	040/14	040/16	040/17
Visibility	9	9	9	8	8	7	7	6	7

Valid	9 hr Thu 09/15Z 09/10L	10 hr Thu 09/16Z 09/11L	11 hr Thu 09/17Z 09/12L	12 hr Thu 09/18Z 09/13L	13 hr Thu 09/19Z 09/14L	14 hr Thu 09/20Z 09/15L	15 hr Thu 09/21Z 09/16L	16 hr Thu 09/22Z 09/17L	17 hr Thu 09/23Z 09/18L	18 hr Fri 10/00Z 09/19L
Temp °F	76°F	78°F	81°F	83°F	84°F	85°F	85°F	84°F	82°F	80°F
Temp °C	24°C	26°C	27°C	28°C	29°C	29°C	29°C	29°C	28°C	27°C
DP Temp °F	67	67	67	66	65	64	63	62	62	63
DP Temp °C	20	20	19	19	18	18	17	17	17	17
Humidity	75%	69%	63%	58%	54%	50%	48%	48%	51%	56%
Wind Chill	76°F	78°F	81°F	83°F	84°F	85°F	85°F	84°F	82°F	80°F
Heat Index	77	81	83	85	86	86	85	85	83	82
FITS	90	92	94	95	96	96	96	95	94	92
Gen Wx	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon	Coming Soon
Cloud Coverage	1	0	0	0	0	0	0	0	0	1
Lowest Ceiling										
Winds (kt)	030/12 22	030/11 20	030/10 20	030/10 19	020/10 19	030/10 19	030/10 19	020/10 18	020/9 16	020/9 15
Cross Wind (kt)	0	0	1	1	1	1	1	1	1	2
1,000 ft MSL Winds	050/17	040/16	040/15	030/15	030/15	030/15	030/15	030/15	030/16	030/18
2,000 ft MSL Winds	040/17	040/18	040/18	040/16	030/15	030/15	030/15	030/15	030/16	040/18
3,000 ft MSL Winds	030/17	030/18	030/19	040/18	030/16	030/15	030/15	030/15	030/16	040/17
4,000 ft MSL Winds	030/18	030/18	030/19	030/18	030/17	030/16	030/15	030/15	040/16	040/17
5,000 ft MSL Winds	030/18	030/19	020/18	020/18	020/17	020/16	020/16	030/15	040/16	050/16
6,000 ft MSL Winds	030/18	030/18	020/17	020/17	020/17	020/17	020/18	020/17	030/17	040/17
Visibility	6	7	9	15	15	15	15	15	15	15

Exercise / Practice: TARP Model

PC: Model Forecasting